

A novel Inverter Topology on Mitigation of Commutation Failure in LCC-Based HVDC Systems

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Abstract—With more applications and larger transmission capacity of LCC-HVDC transmission systems, commutation failure has attracted more attention. In view of this problem, existing methods pay less attention to commutation failures caused by severe short-circuit faults on AC or DC side. Thus an auxiliary method is needed as a supplement. This paper develops a method for SFCLs to suppress commutation failure in HVDC transmission system. A strategy based on SFCLs is designed as an auxiliary method to mitigate commutation failures caused by severe short-circuit faults. The proposed method is based on the proposed module. The operating principles are presented. The performance of the proposed method is validated by simulation results in PSCAD. The simulation results show that the proposed method can effectively prevent the first commutation failure caused by severe short circuit fault.

Keywords—Commutation failure, Superconducting coil, LCC-based HVDC, Serious fault

I. INTRODUCTION

High voltage direct current (HVDC) transmission based on line-commutated converter (LCC) technology has been widely utilized around the world for power transmission. However, some well-known problems associated with LCC-HVDC still exist today which limits the further application of this technology. The most notable one is commutation failure. Commuting failure is one of the common faults of thyristor inverters [1][2]. Although the frequency of commutation failure varies from year to year, there is an obvious upward trend on the whole. The main causes of commutation failure include AC short circuit fault, the increase of non-fault pole current due to electromagnetic coupling between DC lines after the single pole to ground short circuit fault in the bipolar DC system, and the valve fault which is mainly caused by triggering abnormality. 86% of commutation failures were due to AC voltage sags caused by AC short circuit faults, of which 41% were serious faults with AC voltage sags of more than 50% or even 100% before commutation failure [3].

The commutation failure can cause a temporary cessation of power transfer and overheating of the valves. A sudden increase of power transfer between adjacent AC transmission lines can also happen which might affect system transient

stability [4]. The first commutation failure is one of the inducements of the cascading failures in the power system [5]. This puts forward higher requirements for the mitigation strategy of commutation failure: the strategy should be taken as an important part of the measures to prevent the cascading failures in the power system. It can reduce the probability of commutation failures and the impact on the power system after the occurrence of commutation failures. The key of reducing the impact is to limit the DC short-circuit current, to prevent the occurrence of continuous commutation failure, which can avoid the abnormal action of the DC and AC lines.

For the controller modification methods, it has been identified in literature that commutation failures cannot be completely avoided if the fault is electrically close to the inverter [6]. Thus, the objectives of these modifications are either to reduce the probability of commutation failure or to increase the recovery speed of DC system after commutation failure[7]–[9], [10]. The most used method is to increase the firing angle at inverter side immediately after detection of AC voltage disturbance in order to give a larger commutation margin [11]. In [11], AC voltage at inverter side is taken as the commutation failure criterion and the compensation value of the firing angle is output in proportion to the voltage sag, but the compensation is slow when the AC voltage is near the zero-crossing point. In [12][13], there is another possibility to reduce the commutated DC current by lowering the current order at rectifier side upon detection of AC voltage disturbance simultaneously, but the change of current order at rectifier side might not be fast enough given the normally long distance of the DC link. Mirsaeidi S, et al. [14] propose a predictive control strategy, deploying a commutation failure prevention module to mitigate the commutation failures during ac system faults. However, the mitigation of commutation failure caused by low impedance ground is barely effective. Adjusting the control strategy of the converter station has obvious advantages in economy and convenience. It is currently the main method to suppress commutation failure. However, due to the limitation of the adjustment range and adjustment speed of the firing angle, adjusting the control strategy of the converter station has limited effect on commutation failure caused by serious short-

circuit faults. The method cannot cover all commutation failure scenarios.

Another method to suppress commutation failure is to connect additional devices in series. This method is mainly proposed to deal with commutation failure caused by severe short-circuit faults. The more common ones are series capacitors, which constitute a capacitor-commutated converter (CCC) [15].[16] designed a controllable capacitor device based on an anti-parallel thyristor full-bridge sub-module to compensate the commutation voltage, which is called an enhanced capacitor-commutated converter (ECCC), which can effectively suppress the commutation failure caused by the AC short-circuit fault. Compared with CCC, the number of capacitors in ECCC is reduced by half. The capacitor voltage is controllable and coordinated control can be performed, which can improve the system recovery characteristics and reduce the probability of subsequent commutation failure during recovery. While having the same auxiliary commutation effect of CCC, No more relying on fast failure detection. However, the compensation effect of the ECCC on the commutation voltage is limited by the size and rated voltage of the series capacitor. The use of thyristors makes the ECCC unable to actively control. [17][18] proposed a series voltage commutation converter (SVCC) based on series IGBT full-bridge sub-modules, which can dynamically adjust the output voltage to cope with different commutation bus voltage drop degrees. SVCC changes the current source characteristics of LCC in principle. But the current capacity of IGBTs is not as good as that of thyristors. This limits the DC transmission power, increases the risk of overcurrent damage to series devices. The principle of the above series capacitors is basically the same. It is generally composed of a capacitor and a half-bridge/full-bridge circuit composed of half-controlled/full-controlled power electronic devices. It is installed on the AC side of the inverter. When a fault occurs, the capacitor is switched to compensate for commutation. This method requires repeated switching of capacitors during the normal operation of the system to assist the commutation process and ensure that the capacitor voltage is maintained within the set range, which has a certain impact on the normal operation of the system.

Another way to connect additional devices in series is to connect resistors in series. The series resistor can make up for the deficiencies caused by the limited firing angle adjustment range and speed, help to suppress the commutation failure caused by serious fault. The series resistor avoids repeated switching during steady-state operation, reducing the influence on the normal operation of the system. In addition, resistors can be applied on the DC line to suppress the commutation failure caused by the DC fault. [19] proposed a commutation failure suppression device with IGBT and resistor in parallel. When a fault on the AC side is detected, the device controls the IGBT to disconnect and the resistor is put in to limit the short-circuit current and reduce the voltage drop. However, since the active power is generated when the resistor is connected in series in the line, a fully controlled device or circuit breaker must be connected in parallel to control the resistor to be put into the circuit only after the fault occurs. The action time depends on the fault detection time.

In recent years, with the breakthrough of superconducting materials, the method of using superconducting fault current limiter (SFCL) to suppress commutation failure has gained some attention. In addition to the advantages of resistors,

SFCLs can be kept in series in the system line during steady-state operation. There is no need to connect full-control devices in parallel. SFCLs generate resistance after a fault occurs, so the operating time does not depend on the fault detection. Therefore, it has certain advantages to use the superconducting current limiter-based suppression method as an auxiliary method to adjust the inverter strategy.

This paper is organized as follows. In Section 2, the proposed method on mitigation of commutation failure based on SFCLs is described. Section 3 explains the principles of SFCL and operation of the proposed method. In Section 4, the effectiveness of the proposed method is validated by various simulations in PSCAD. Finally, Section 5 concludes the paper.

II. THE INVERTER WITH SFCLs

Fig. 1 shows the proposed converter circuit configuration and the connected AC system at inverter side, where VT1-VT6, VTB and VTF are thyristor valves. SupDC and SupBri are the superconducting modules. Zinv and Zfil are the equivalent impedances for AC system on the inverter side and filters. SupCoil is superconducting coils. The whole system is based on an original CIGRE Benchmark 12-pulse LCC-HVDC system with AC systems at both ends rated in 50 Hz. Two 6-pulse bridges are connected in series and are connected to the AC system through transformers.

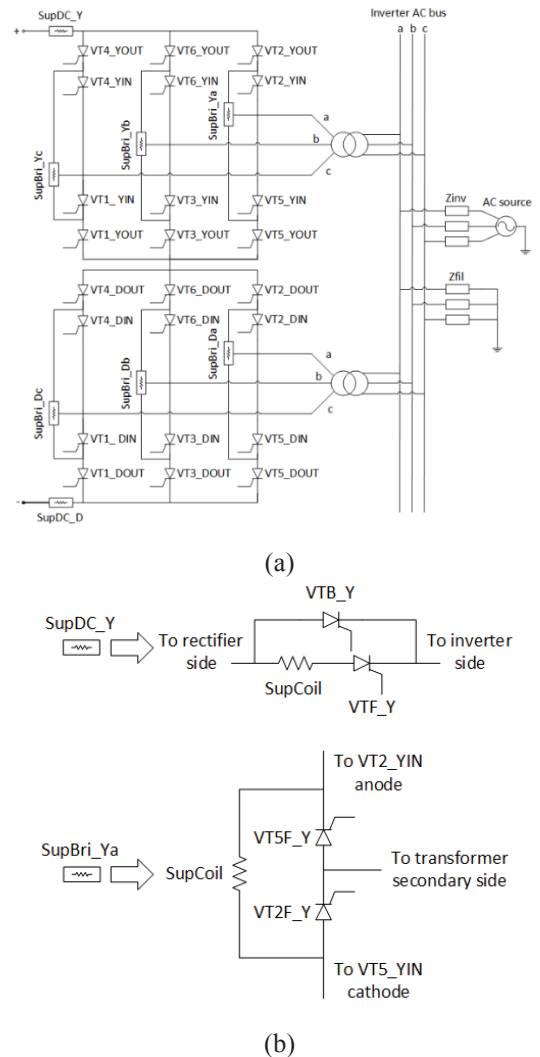


Fig. 1. The proposed inverter topology and connected AC system. (a) System configuration; (b) Superconducting module configuration.

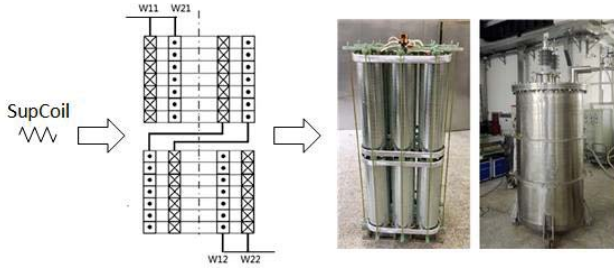


Fig. 2. Superconducting coil.

The only additions to the original system are the superconducting modules which are connected to DC lines and bridges. These superconducting modules are named as superconducting DC modules or superconducting bridge modules because of different locations. There are superconducting coils in superconducting modules shown in Fig. 1. Each superconducting coil can be realized by several series and parallel connected sub-coils to achieve higher insertion resistance and higher critical current. In the following modeling, single superconducting coil is assumed for the sake of demonstration of the approach. One superconducting DC module is needed for each pole and one superconducting bridge module is needed for each phase. Thus, 8 modules are needed for the 12-pulse HVDC system. It can be seen from Fig. 1 that each module consists of two thyristor-valves. In a superconducting DC module, the left-hand side is connected to rectifier side while the right-hand side is connected to inverter side. In a superconducting bridge module, the up-hand side and the down-hand side are connected to the up-bridge and the down-bridge respectively while the middle-hand side is connected to the transformer secondary winding.

The superconducting coil shown in Fig. 2 is constructed by a solenoidal configuration. The superconducting coil consists of 2 inner and 2 outer cylinders. The 4 cylinders are arranged coaxially. 1 outer cylinder and 1 inner cylinder form a group. The inner cylinder in the upper group is connected in series with the outer cylinder in the lower group and the outer cylinder in the upper group is connected in series with the inner cylinder in the lower group to ensure that the same amount of current through the inner and outer cylinders in each group. The inner and outer cylinders of each group are spiraled in opposite directions to ensure that the magnetic fields cancel each other out so that the superconducting coils are resistive. The measured leakage inductance of a superconducting coil is only $3.7 \mu\text{H}$ when the difference in radius between the inner and outer cylinders is 10 mm [20]. The skeleton of the cylinder is made of epoxy resin and the outer layer of the skeleton is slotted. Superconducting tapes are wound around it. In this paper, an $\text{Y2O}_3\text{-BaO-CuO}$ (YBCO) tape called AMSC8602 is chosen. Several superconducting coils are assembled in series and parallel and then placed into a Dewar and connected to refrigeration system to ensure that superconducting coils are immersed in liquid nitrogen.

Superconducting coils exhibit two states: the superconducting state and the normal state, which is shown in Fig. 3. When the current through the superconducting coil is less than its critical current, the coil is in the superconducting state and its resistance is zero. When the current through the superconducting coil is greater than its critical current, the coil quenches. Then the coil is in the normal state and has resistance. When the current is cut off and after cooling, the

superconducting coil returns to the superconducting state. In the superconducting DC module shown in Fig. 1, the superconducting coil will be inserted when VTF_Y is switched on and VTB_Y is switched off. The superconducting coil will be bypassed when VTB_Y is switched on and VTF_Y is switched off. In the superconducting bridge module shown in Fig. 1, the superconducting coil will be inserted when VT2F_Y and VT5F_Y are switched on, meanwhile, VT2_Y and VT5_Y are switched off. The superconducting coil will be bypassed when VT2_Y and VT5_Y are switched on, meanwhile, VT2F_Y and VT5F_Y are switched off. To enable comparisons with the original system, the converter control system at both ends is kept the same as the original system.

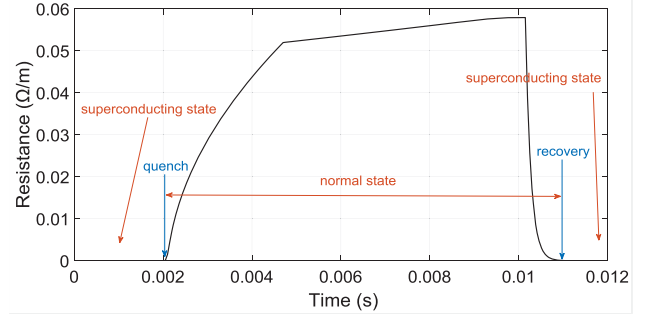


Fig. 3. Resistance of superconducting coils under an impact.

III. CONTROL STRATEGIES

A. Commutation failure

The essence of commutation failures is that the valve in the outgoing phase fails to be switched off because its current does not drop to zero in a required time. Theoretical analysis of commutation failures is presented in [1], which concludes that the inverter AC voltage magnitude reduction is the most dominant factor for commutation failure under both single-phase and three-phase fault. Voltage phase angle shift is also the reason for the onset of commutation failure. Another important conclusion is that the DC current increase caused by single-pole fault and electromagnetic coupling between DC lines [21]. Valve faults can also cause commutation failures. It is due to fact that the control system will not be able to eliminate commutation failures caused by serious faults [14]. However if the aim is to eliminate commutation failures rather than reduce its probability, the most direct way is to reduce the current of the outgoing phase. In this paper, SFCLs are adopted to eliminate the commutation failures under zero impedance single-phase and three-phase faults at inverter AC bus, zero impedance single-pole faults at DC lines and valve faults. The usefulness of the method then largely depends on the control strategy and selection of superconducting coil resistance which are described in following sections. In a commutation, the outgoing phase and the incoming phase are shown in Fig. 4.

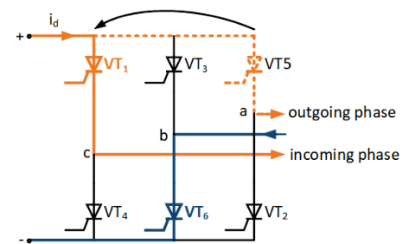


Fig. 4. The outgoing phase and the incoming phase in a commutation.

B. Operating principles

The control strategy is called ‘flow’ & ‘bypass’ method and the current path under each method is shown in Fig. 5. If faults occur in the outgoing phase, superconducting modules will keep in series in the circuit. If faults occur in the incoming phase, superconducting modules will be bypassed.

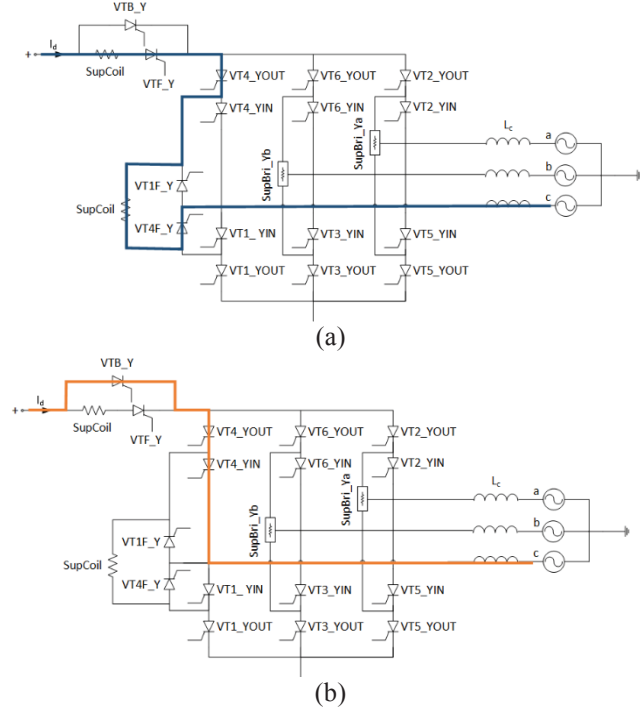


Fig. 5. ‘Flow’ & ‘Bypass’ method of SFCL control. (a) The flow state; (b) The bypass state.

Through the parallel connection of several superconducting coils, the critical current of the superconducting module is appropriately higher than the rated current of the line where it is located. The superconducting module usually shows zero resistance, which does not affect the normal operation of the line and the inverter. When a serious fault occurs, the line current rises rapidly and the superconducting coil automatically quenches. The superconducting module maintains the flow or bypass state according to the control logic. Refer to Section 4 for the definition of serious faults and minimum limiting resistance. The action process of the superconducting module can be divided into 4 stages.

Stage I: Before the short-circuit fault occurs, the line is operating normally. The superconducting module is in the flow state. At this time, the line current is lower than the critical current of the superconducting module. The superconducting coil is in the superconducting state, which shows zero resistance. The voltage sensor of the superconducting module monitors the voltage at both ends of superconducting coils to determine whether it quenches.

Stage II: When a short circuit fault occurs, the line current exceeds the critical current of the superconducting module. The superconducting coil quenches. The voltage at both ends of the superconducting module exceeds the threshold, then the superconducting module controller judges the line type and fault type. If the line type is AC side incoming phase or the fault type is non-serious fault, the superconducting module should be in the bypass state. Otherwise, the superconducting module should be in the flow state.

Stage III: If the superconducting module is in the flow state after Stage II, the controller judges the line type and the fault type again, i.e., repeats Stage II.

Stage IV: If the superconducting module is in the bypass state after Stage II, it is determined that the action process is over when the voltage sensor of the superconducting module detects that the voltage at both ends of superconducting coils drops below the threshold. The superconducting module should be in the flow state and return to Stage I.

The control logic of superconducting modules is shown in Fig. 6.

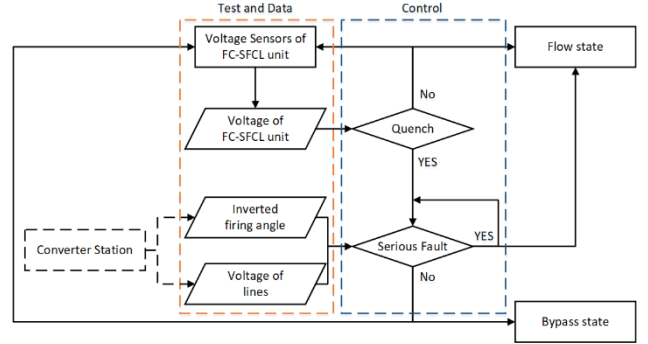


Fig. 6. Operating principles of superconducting modules.

The advantage of SFCLs is that in the steady state, superconducting coils are in the superconducting state, which does not affect the operation of the system. The superconducting module can always operate in the flow state when there is no fault. When there is a short-circuit fault in the line, the superconducting module quenches because the operating current exceeds the critical current, which limits the short-circuit current. After the superconducting module in the bypass state returns to the superconducting state, a trigger signal is sent to the thyristor and the superconducting module is re-inserted into the line and returns to the flow state.

Fig. 7 shows the control logic for switching from the bypass state to the flow state. Measure the voltage across superconducting coils in the bypass state. If the measured value is lower than the threshold value, it is considered that the superconducting module has completed the quench recovery. The state of the module changes from the bypass state to the flow state. If the voltage measurement value is higher than the threshold value, it is considered that coils are still in the normal state, the superconducting module remains in the bypass state and continues to measure its voltage.

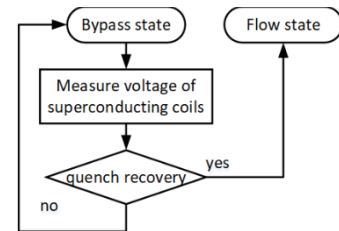


Fig. 7. Control logic from bypass state to flow state.

The superconducting module is in flow state before the fault occurs and at the initial stage of the fault. Only when the fault occurs and the superconducting module quenches, the controller will start to judge the fault type, then bypass the superconducting module which should not be in the flow state.

and has quenched. So requirements for the fault judgment time are not too high.

There are two scenarios in which the superconducting module needs to be bypassed in time: the fault line in the single-pole grounding short-circuit fault of the bipolar HVDC transmission system and the incoming phase in the AC short-circuit fault. Except for these two situations, keeping the superconducting module in the flow state is beneficial to suppress the commutation failure. Therefore, the key point of the control logic is how to transfer the superconducting module which has quenched but not needed from the flow state to the bypass state during the fault. In order to suppress commutation failures effectively, it should be prudent to judge that the superconducting module changes from the flow state to the bypass state. The judgment principle of this process is that assuming that there is no superconducting module, whether the HVDC transmission system will occur commutation failure, that is, to judge according to

$$\cos \gamma_{\min} > \cos \beta + \frac{2\omega L i_d}{u_{\Delta f}} \quad 1$$

where $\gamma_{\min}$, ω , L are constants, i_d , $u_{\Delta f}$ represent the instantaneous values of the DC current and the commutation voltage, respectively.

Fig. 8 shows the control logic from the flow state to the bypass state. The blue and red dotted boxes are the specific control processes of the superconducting bridge module and the superconducting DC module, respectively. The other parts are the control processes common to all modules. The method of judging whether the quench bridge module is on the incoming phase is to judge whether the latest trigger signal before the fault is sent to the thyristor of the phase. If it is sent to the phase, the corresponding phase is the incoming phase or is not in the commutation process. It is ensured that the phase must not be the outgoing phase and bypassing the superconducting module will not have a negative impact on suppressing the commutation failure. If the last trigger signal is not sent to this phase, then this phase must not be the incoming phase. So maintaining the flow state of the superconducting module will not deteriorate the commutation process.

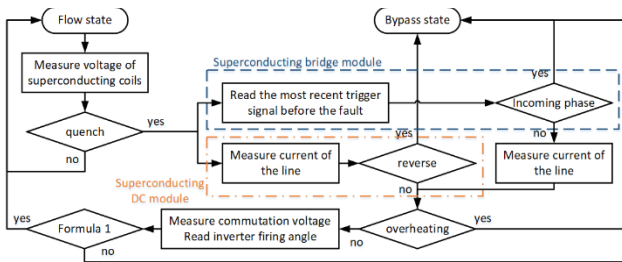


Fig. 8. Control logic from flow state to bypass state.

The method of judging whether the quench DC module is on the fault line is to judge whether the current of the DC line where it is located is reversed. In a DC line, only the inverter side current on the fault line will reverse in the event of a single-pole grounding short-circuit fault, while the non-fault line inverter side current will not reverse under no fault or any fault condition. This part of the judgment can achieve fast action.

After this part of the judgment, it is necessary to calculate the resistance of superconducting coils to determine whether superconducting coils is overheated and bypass them if it is damaged by overheating. Finally, measure the commutation voltage, read the inverter firing angle and judge whether the superconducting module needs to be kept in the flow state by formula 1.

IV. SIMULATION RESULTS

Based on the CIGRE Benchmark model, the proposed method is built in PSCAD. The topology of the CIGRE Benchmark model is shown in Fig. 9. The simulation model of SFCL coils is shown in [22].

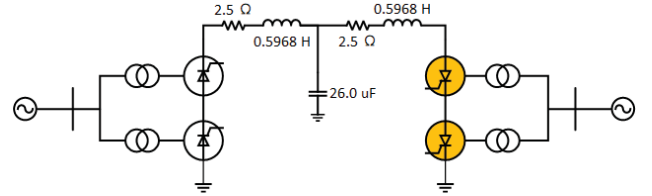


Fig. 9. The structure of CIGRE Benchmark model.

In order to verify the effectiveness of the proposed method in suppressing commutation failures caused by AC short-circuit faults, SLG faults and three-phase (TP) short-circuit faults with different fault resistances and different fault times are performed on the inverter side. The result is shown in Fig. 10. In the figure, green points indicate that the original control strategy effectively suppresses the first commutation failure; blue points indicate that the original control strategy fails to suppress the first commutation failure, but suppress subsequent commutation failures to prevent DC system blocking; red points indicate that the original control strategy fails and DC system blocking has occurred, but the proposed method can effectively suppress the first commutation failure; purple points indicates the proposed method fails to suppress commutation failures. The results show that, except for the zero-impedance TP faults, the proposed method can effectively suppress the first commutation failure.

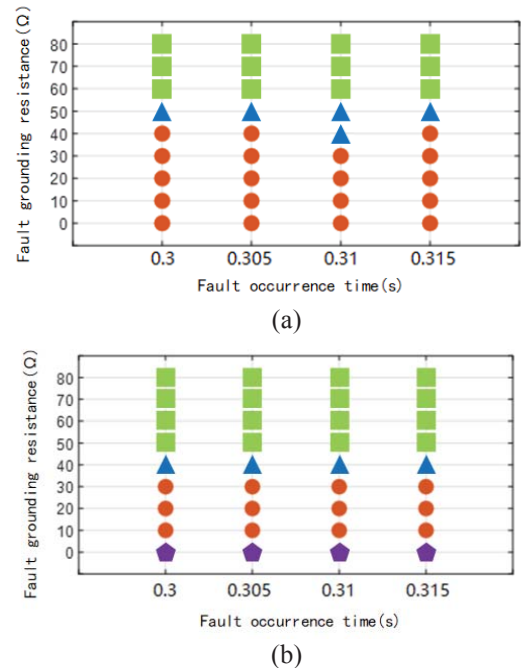


Fig. 10. The commutation of the bipolar HVDC under different fault conditions. (a) SLG fault; (b) TP fault.

Fig. 11 shows simulation results when a zero-impedance single-phase grounding fault occurs on the AC bus on the inverter side. In the figure, (a) is the AC current situation under the existing control strategy of CIGRE Benchmark converter station, (b) is the AC current situation after adopting the proposed strategy in this paper, (c) is the DC current situation under the existing strategy and the proposed strategy, (d) is the trigger signal of the bypass thyristor valve in the superconducting module, (e) is the resistance of superconducting module. When the existing strategy is adopted, the AC short-circuit fault causes the AC and DC currents to increase, which in turn causes the first commutation failure and leads to subsequent commutation failures. When the strategy proposed in this paper is adopted, the superconducting module quickly quenches in the initial stage of the AC fault, which limits the fault current within a reasonable range and effectively avoids the occurrence of the first commutation failure.

It can be seen from the figure that the superconducting module remains in the flow state before the fault occurs. When the fault occurs, the superconducting module quenches. But according to the control logic, the controller determines that the fault phase is the incoming phase. So the superconducting module is bypassed. Due to the timely bypass, the resistance of the superconducting module is small and the quench recovery is relatively fast. After the controller determines that the superconducting module has completed the quench recovery. Then the superconducting module turns into flow state. At this time, the fault phase current is still greater than the critical current. The superconducting module quenches and the controller determines that the phase is not the incoming phase. The superconducting module remains in the flow state, limiting the short-circuit current, suppressing commutation failure. At 0.307 s, the superconducting module is still in the flow state and quenches, but the fault phase becomes the incoming phase again.

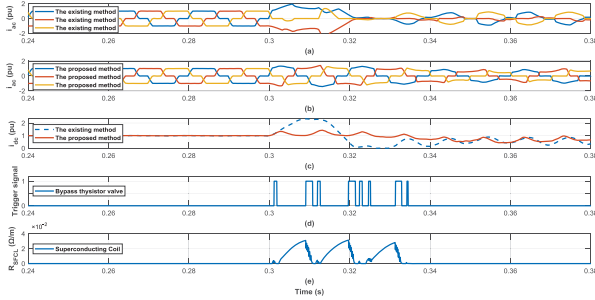


Fig. 11. Results under an SLG fault (fault grounding resistance is 0).

Fig. 12 shows the simulation results when the AC bus on the inverter side has a single-phase grounding fault with a fault resistance of 50 Ω .

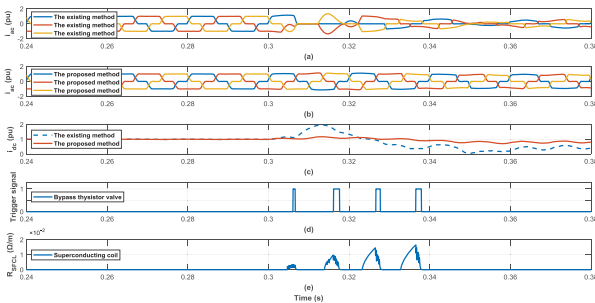


Fig. 12. Results under an SLG fault (fault grounding resistance is 50 Ω).

Fig. 13 shows the simulation results when a zero-impedance three-phase grounding fault occurs on the AC bus on the inverter side. The duration of this fault state is 20 ms. When the existing suppression strategy is adopted, the AC short-circuit fault causes the AC and DC currents to increase, which in turn causes the first commutation failure and the subsequent commutation failure. The output power of the DC line is almost reduced to zero. When the strategy proposed in this paper is adopted, the superconducting module quickly quenches at the initial stage of the AC fault, limiting the fault current to a reasonable range. Although the first commutation failure cannot be avoided, subsequent commutation failure is effectively avoided.

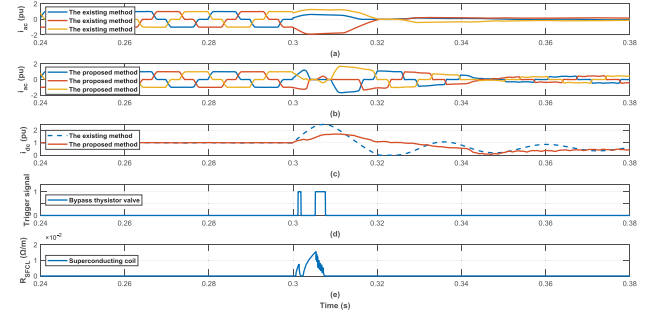


Fig. 13. Results under a TP fault (fault grounding resistance is 0).

Fig. 14 shows the simulation results when the AC bus on the inverter side has a three-phase grounding fault with a fault resistance of 40 Ω .

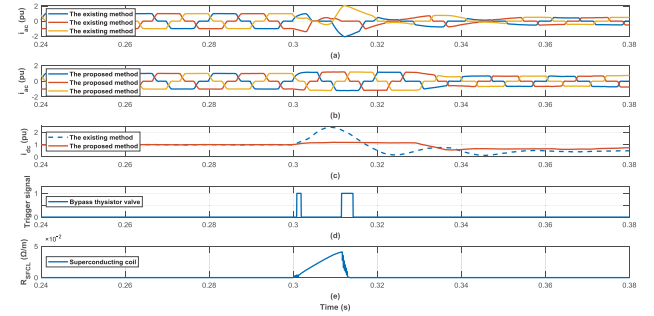


Fig. 14. Results under a TP fault (fault grounding resistance is 40 Ω).

In order to verify the effect of the proposed method on the suppression of commutation failures caused by SPG faults, a true bipolar DC system model based on the CIGRE Benchmark model is built. The topology is shown in Fig. 15.

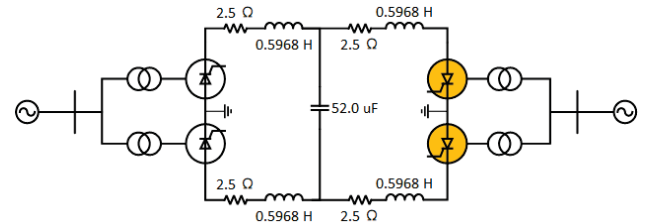


Fig. 15. The structure of the true bipolar HVDC system based on CIGRE Benchmark model.

Based on the true bipolar HVDC system model, the simulation verifies the mitigation of commutation failures under the proposed method. SPG faults are carried out at 0.3 s. The fault duration is 100 ms and the fault resistance is 0. The fault occurs at the DC outport of the inverter. Appropriately decrease the operating power of CIGRE Benchmark model, the results are shown in Fig. 16. The

superconducting modules limits the current and effectively suppresses the first commutation failure.

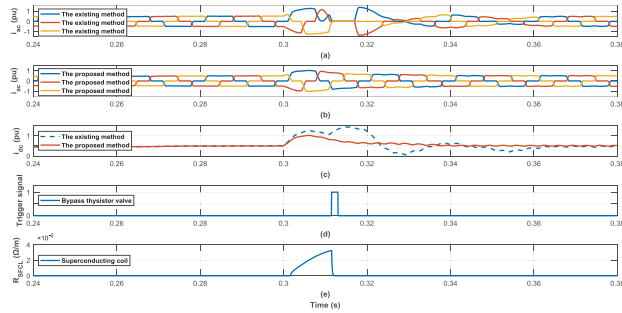


Fig. 16. Results under an SPG fault (half of the rated power).

V. CONCLUSION

This paper proposes a superconducting fault current limiting converter topology and its control logic. Simulation models based on the CIGRE Benchmark model are built in PSCAD to verify the effectiveness of the proposed method in suppressing commutation failures caused by serious faults.

The simulation results show that the proposed method can effectively suppress the first commutation failure caused by single-phase grounding faults, three-phase faults and single-pole grounding short-circuit faults in the bipolar DC system. The proposed method can also effectively suppress subsequent commutation failures caused by thyristor valve controller faults. Its inhibition ability and effect are better than original strategies.

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